

USN

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Regular Examinations Sep/Oct – 2024
Electronics and Communication Engineering

TRANSMISSION LINES AND ELECTROMAGNETIC FIELDS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

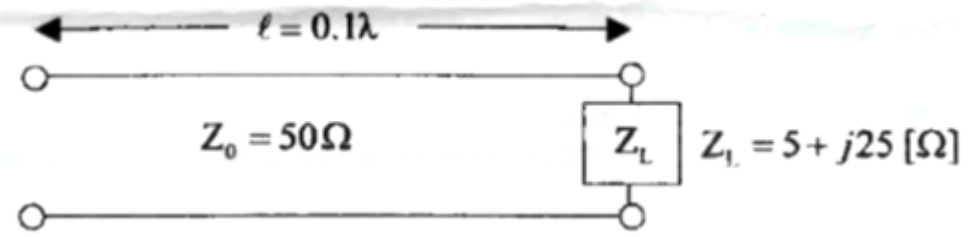
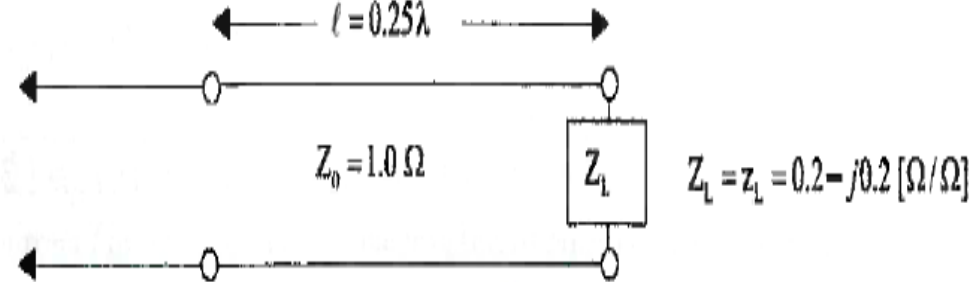
PART-A

M BT CO

1	1.1	Give four analogies between Telegraph's equation and Helmholtz's Wave equation.	02	1	1
	1.2	A transmission line of characteristic impedance $500\angle -43^\circ \Omega$ and propagation constant $0.07 + j0.08$ per km is terminated properly at the load end. A voltage of $5V$ is applied at sending end. Find Voltage and current at a distance $10km$ from sending end?	02	2	1
	1.3	A 100Ω lossless line connects a signal of $100kHz$ to a load of 140Ω . The load power is $100mW$. Calculate Reflection coefficient and $VSWR$.	02	2	2
	1.4	A 300Ω lossless line is terminated by a load of 78Ω operating at $40MHz$. What type of stub is suitable for this line? Find length of the stub.	02	2	2
	1.5	If $\mathbf{D} = 2r \hat{r} C/m^2$, find the total electric flux leaving the surface of the cube, $0 < x, y, z < 0.4$	02	2	3
	1.6	Planar current sheets of $K = 30\hat{z} A/m$ and $-30\hat{z} A/m$ are located in free space at $x = 0.2$ and $x = -0.2$ respectively. For the region $-0.2 < x < 0.2$. Find $\mathbf{H}$.	02	2	3
	1.7	Give four analogies between Electrostatics and Magnetism.	02	1	3
	1.8	With necessary equation explain Scalar and Vector Magnetic potential.	02	1	4
	1.9	Write Maxwell's equation in Integral form.	02	1	4
	1.10	With necessary equation explain Skin effect in a good conductor.	02	1	4

PART-B

2	a	An open wire transmission line having primary constants $R \text{ ohm/m}$, $L \text{ H/m}$, $G \text{ mho/m}$ and $C \text{ F/m}$ operating at $f \text{ Hz}$ then: i) Derive an expression for transmission line using distributed parameter model ii) If $V = V_s$ and $I = I_s$ at generator end Derive an expression to determine Voltage and current at any point 'l' along the line.	08	2	1
---	---	--	----	---	---

b	The open circuit and short circuit impedance at $8000/2\pi$ Hz of a 1 km length transmission line are respectively $Z_{oc} = 2500\angle -70^\circ \Omega$ and $Z_{sc} = 49\angle 25^\circ \Omega$. i) Determine the characteristic impedance, Propagation constant, Phase constant and the attenuation constant of the line ii) Determine Primary constants R, L, G, and C of the line.	08	3	1
3 a	The 0.1λ length line shown in Fig. 3.a has a characteristic impedance of 50Ω and is terminated with a load impedance of $Z_L = 5 + j25\Omega$. Find:  Fig. 3.a i) The impedance at $l = 0.1\lambda$ ii) The VSWR on the line? iii) Γ_L iv) Γ at $l = 0.1\lambda$ from the load?	08	3	1
b	A point charge 100pC is located at $(4, 1, -3)$ while the x-axis carries charge 2nC/m. If the plane $z = 3$ also carries charge 5 nC/m^2, find E at $(1, 1, 1)$ OR	08	3	1
4 a	A transmission line has $Z_0 = 1.0$, $Z_L = 0.2 - j0.2 \Omega$ as shown in Fig. 4.a. Using Smith Chart, Find:  Fig. 4.a i) Impedance at 0.25λ ii) The VSWR on the line iii) How far from the load is the first voltage minimum?	08	3	2
b	Three concentric spherical shells $r = 1$, $r = 2$, and $r = 3\text{m}$ respectively, have charge distributions 2, -4, and $5\mu\text{C/m}^2$ i) Calculate the flux through $r = 1.5\text{m}$ and $r = 2.5\text{m}$ ii) Find D at $r = 0.5$, $r = 2.5$, and $r = 3.5\text{m}$	08	3	2
5 a	A spherical charge distribution: $\rho_v = \begin{cases} \rho_0 \frac{r}{a}, & r \leq a \\ 0, & r > a \end{cases}$ Find: i) E and V for $r \leq a$ ii) E and V for $r > a$	08	4	2

	b In a vacuum diode. The space charge density is given by: $\rho_v = \frac{4\epsilon_0 V_0}{9s^{4/3}x^{2/3}}$ Where V_0 is the voltage at the anode (the voltage at the cathode is zero), s is the distance between the electrodes; and x is the distance of a point to the cathode. Given at $x = 0$, $V = 0$ and $x = s$, $V = V_0$, find: i) V as a function of x ii) E as function of x OR	08	3	2
6	a The surface $x = 0$ separates two perfect dielectrics. For $x > 0$, $\epsilon_1 = 3$, while $\epsilon_2 = 5$ for $x < 0$. If $E_1 = 80\hat{x} - 60\hat{y} - 30\hat{z} \text{ V/m}$, find: i) D_2 ii) E_2 iii) The angles E_1 and E_2 make with the interface iv) Energy densities in each medium b Conducting spherical shells with radii 'a' and 'b' are maintained at a potential difference of V_0 such that $V(r = b) = 0$ and $V(r = a) = V_0$. Determine: i) E in the region between the shells using Laplace Equation ii) V in the region between the shells using Laplace Equation iii) The surface charge density on the shells and iv) The capacitance of the capacitor.	08	3	2
7	a The cylindrical shell, $2\text{mm} < \rho < 3\text{mm}$, carries a uniformly – distributed total current of 8A in the $-\hat{z}$ direction, and a filament on the z axis carries 8A in the $\hat{z}$ direction. Find H at i) $\rho < 2\text{mm}$ ii) $2\text{mm} < \rho < 3\text{mm}$ iii) $\rho > 3\text{mm}$ iv) Also compute Magnetic flux on surface $\phi = 0$, $\rho \leq 2\text{mm}$, $0 \leq z \leq 1\text{m}$ b Show that Normal component of Magnetic flux is continuous and tangential component of Magnetic field is continuous for two magnetic media μ_1 and μ_2. OR	08	3	3
8	a In a long coaxial transmission line, the radius of inner conductor is 'a' and carries current I in $+z$ direction. The inner radius of outer conductor is 'b' and has a thickness of 't' and carries current I is carried in $-z$ direction. Find magnetic field density at i) $\rho \leq a$ ii) $a \leq \rho \leq b$ iii) $b \leq \rho \leq (b + t)$ iv) $\rho > b$ b Given $A = 50\rho^2\hat{z} \text{ Wb/m}$ in a certain region of free space: i) Find H and B ii) Find J: iii) Use J to find the total current crossing the surface $0 \leq \rho \leq 1$, $0 \leq \phi < 2\pi$, $z = 0$ iv) Use the value of $H\phi$ at $\rho = 1$ to calculate I for $\rho = 1$, $z = 0$	08	3	3

9	a i) Stating the limitation of Ampere circuital law for steady currents ii) Derive Equation of continuity iii) Using the result in ii) show that $\nabla \times H = J + \frac{\partial D}{\partial t}$ iv) State Faraday's law of electromagnetic induction. Show how it leads to the Maxwell equation $\nabla \times E = -\frac{\partial B}{\partial t}$	08	2	4
	b A uniform plane wave in air is normally incident on an infinite dielectric material having $\epsilon = 3\epsilon_0$ and $\mu = \mu_0$. If the incident wave is $E_i = 10 \cos(\omega t - z)\hat{y}$ V/m, find: i) λ and μ of the wave in air and the transmitted wave in the dielectric medium ii) The incident Magnetic field iii) Γ and τ iv) The total electric field and the time-average power in both regions.	08	3	4
OR				
10	a A plane wave propagating along the +z direction is incident normally on the boundary $z = 0$ between medium 1 ($z < 0$) characterized by $\sigma_1, \mu_1, \epsilon_1$ and medium 2 ($z > 0$) characterized by $\sigma_2, \mu_2, \epsilon_2$. An incident wave ($E_i, H_i$) in medium 1 is traveling along $\hat{z}$ with propagation constant γ_1 and characteristic impedance η_1 $E_i = E_{io} e^{-\gamma_1 z} \hat{x}$ The reflected wave (E_r, H_r) in medium 1 is travelling along $-\hat{z}$ with propagation constant γ_1 and characteristic impedance η_1 $E_r = E_{ro} e^{\gamma_1 z} \hat{x}$ The transmitted wave (E_t, H_t) in medium 2 is traveling along $\hat{z}$ with propagation constant γ_2 and characteristic impedance η_2 $E_t = E_{to} e^{-\gamma_2 z} \hat{x}$ Derive an expression for i) Reflected coefficient and transmitted coefficient ii) (E_r, H_r) in terms of (E_{io}, H_{io}) iii) (E_t, H_t) in terms of (E_{io}, H_{io}) iv) The standing wave in medium 1	10	3	4
	b The magnetic field component of a plane wave in a lossless dielectric is: $H = 30 \sin(2\pi \times 10^8 t - 5x)\hat{z}$ mA/m i) Find $\mu_r = 1$, find ϵ_r ii) Calculate the wavelength and wave velocity iii) Determine the wave impedance iv) Determine the polarization of the wave v) Find the corresponding electric field component vi) Find the displacement current density.	06	3	4